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Project Title	SQL Query Implementation for E-Commerce Database
Week	Week 7

WEEK-7 PROJECT

Task Name:

SQL Query Implementation for E-Commerce Database

Description:

This Week-7 task focuses on implementing SQL queries for an E-Commerce database. The objective is to retrieve, filter, sort, and analyse customer, product, order, and inventory information using fundamental SQL commands. Students will work with SELECT, WHERE, ORDER BY, DISTINCT and related filtering conditions to answer common business questions and generate basic reports.

The project should demonstrate how SQL can be used to search products by price, category, and availability, retrieve customer and product information, apply multiple filtering conditions, and produce useful business-oriented query results from a structured E-Commerce database.

1. E-COMMERCE DATABASE OVERVIEW

The database should represent a basic online shopping system. It can contain customer, product, category, order, order-item, and inventory information.

- Customers: customer_id, customer_name, email, city, registration_date
- Products: product_id, product_name, category_id, price, stock_quantity, availability
- Categories: category_id, category_name
- Orders: order_id, customer_id, order_date, order_status, total_amount
- Order_Items: order_id, product_id, quantity, unit_price

Example product: Product = Wireless Headphones, Category = Electronics, Price = ₹2,499, Stock = 35, Availability = Available.

2. DATABASE TABLE STRUCTURE

A simple relational structure should be created before writing the SQL queries. Primary keys identify individual records, while foreign keys connect related tables.

Table	Important Columns	Purpose
Customers	customer_id, name, email, city	Stores customer information
Categories	category_id, category_name	Stores product categories
Products	product_id, name, category_id, price, stock, availability	Stores product details
Orders	order_id, customer_id, order_date, status, total_amount	Stores order information
Order_Items	order_id, product_id, quantity, unit_price	Stores products within orders

Sample product records

product_id	product_name	category	price	stock	availability
P101	Wireless Headphones	Electronics	2499	35	Available
P102	Mechanical Keyboard	Electronics	3499	12	Available
P103	Running Shoes	Footwear	1999	0	Out of Stock
P104	Backpack	Bags	1299	25	Available

Primary SQL objective

The student should use the database tables to answer practical E-Commerce questions through SQL queries instead of manually filtering the data.

3. SELECT QUERY IMPLEMENTATION

The SELECT statement is used to retrieve specific columns or all columns from a table. It forms the foundation of the E-Commerce query implementation.

Retrieve all products

```
SELECT *  
FROM Products;
```

Retrieve selected product information

```
SELECT product_id, product_name, price, availability  
FROM Products;
```

Retrieve customer information

```
SELECT customer_id, customer_name, email, city  
FROM Customers;
```

Retrieve product names and prices

```
SELECT product_name, price  
FROM Products;
```

The SELECT queries should demonstrate that the student can retrieve only the information required for a particular business question.

4. DISTINCT QUERY

DISTINCT removes duplicate values from the result set. It is useful for identifying unique categories, cities, or availability states.

```
SELECT DISTINCT category_id  
FROM Products;  
SELECT DISTINCT availability  
FROM Products;  
SELECT DISTINCT city  
FROM Customers;
```

5. WHERE CLAUSE AND FILTERING CONDITIONS

The WHERE clause filters records according to a condition. It is one of the most important parts of the project because E-Commerce users and administrators frequently search for records matching specific criteria.

Search products by price

```
SELECT product_name, price
FROM Products
WHERE price < 2000;
SELECT product_name, price
FROM Products
WHERE price BETWEEN 1000 AND 3000;
```

Search products by category

```
SELECT product_name, price
FROM Products
WHERE category_id = 2;
```

Search available products

```
SELECT product_name, stock_quantity
FROM Products
WHERE availability = 'Available';
```

Combine multiple conditions

```
SELECT product_name, price, stock_quantity
FROM Products
WHERE price < 3000
    AND stock_quantity > 0;
```

OR condition

```
SELECT product_name, category_id
FROM Products
WHERE category_id = 1
    OR category_id = 3;
```

Students should demonstrate comparison operators such as =, <, >, <=, >=, and <> together with logical operators such as AND and OR.

6. ORDER BY QUERY IMPLEMENTATION

ORDER BY sorts query results. It can be used to display products from low to high price, high to low price, or customers alphabetically.

Sort products by price in ascending order

```
SELECT product_name, price
FROM Products
ORDER BY price ASC;
```

Sort products by price in descending order

```
SELECT product_name, price
FROM Products
ORDER BY price DESC;
```

Sort customers alphabetically

```
SELECT customer_name, city
FROM Customers
ORDER BY customer_name ASC;
```

Sort using multiple columns

```
SELECT product_name, category_id, price
FROM Products
ORDER BY category_id ASC, price DESC;
```

Business use

- Display the cheapest products.
- Display the highest-priced products.
- Arrange customers alphabetically.
- Sort products within each category by price.
- Prepare ordered results for reports.

7. PRODUCT SEARCH SCENARIOS

The project should include realistic product-search queries based on price, category, stock, and availability.

Requirement	Example SQL condition
Price below ■2,000	WHERE price < 2000
Price between ■1,000 and ■3,000	WHERE price BETWEEN 1000 AND 3000
Available products	WHERE availability = 'Available'
Products with stock	WHERE stock_quantity > 0
Electronics category	WHERE category_id = 1

8. RETRIEVING CUSTOMER AND PRODUCT INFORMATION

Customer and product information can be retrieved independently or together. The project should demonstrate queries that provide useful information to an administrator or business analyst.

Customer search by city

```
SELECT customer_name, email, city
FROM Customers
WHERE city = 'Chennai';
```

Customer search by name

```
SELECT customer_id, customer_name, email
FROM Customers
WHERE customer_name = 'Arun Kumar';
```

Products currently in stock

```
SELECT product_name, price, stock_quantity
FROM Products
WHERE stock_quantity > 0
ORDER BY stock_quantity DESC;
```

Products in a selected price range

```
SELECT product_name, price
FROM Products
WHERE price >= 1500
    AND price <= 5000
ORDER BY price ASC;
```

Using LIKE for search

```
SELECT product_name, price
FROM Products
WHERE product_name LIKE '%phone%';
```

LIKE can be used when the user does not know the complete product name. The wildcard % represents any sequence of characters.

9. BASIC BUSINESS REPORTS

SQL queries can convert raw database records into simple business reports. These reports help identify product availability, price ranges, and customer distribution.

- List all available products.
- Find products below a specified price.
- Find the most expensive products.
- List unique product categories.
- List customers from a particular city.
- Find products that are out of stock.

10. AGGREGATE BUSINESS REPORTS

Basic business reporting can be extended with aggregate functions such as COUNT, SUM, AVG, MIN, and MAX. These functions summarise data instead of returning every individual row.

Count products

```
SELECT COUNT(*) AS total_products
FROM Products;
```

Find average product price

```
SELECT AVG(price) AS average_price
FROM Products;
```

Find minimum and maximum prices

```
SELECT MIN(price) AS lowest_price,
       MAX(price) AS highest_price
FROM Products;
```

Count available products

```
SELECT COUNT(*) AS available_products
FROM Products
WHERE availability = 'Available';
```

Category-wise product count

```
SELECT category_id, COUNT(*) AS product_count
FROM Products
GROUP BY category_id;
```

Business questions answered

- How many products are available?
- What is the average product price?
- What are the minimum and maximum product prices?
- How many products belong to each category?
- How many products are currently out of stock?

11. SAMPLE END-TO-END QUERY WORKFLOW

The project should demonstrate a progression from simple retrieval queries to more useful filtered and business-reporting queries.

E-Commerce Database
↓
Create Tables and Insert Records
↓
SELECT Basic Information
↓
DISTINCT Unique Values
↓
WHERE Filtering
↓
Price / Category / Availability Search
↓
ORDER BY Sorting
↓
Aggregate Functions
↓
Basic Business Reports

12. RECOMMENDED TECHNOLOGIES

- MySQL, PostgreSQL, Oracle SQL, or SQLite for SQL implementation.
- MySQL Workbench, pgAdmin, Oracle SQL Developer, or another SQL client for execution.
- CSV files can be used for initial sample data if required.
- A database export can be used to demonstrate final results.

Query implementation requirements

- Create the required tables with appropriate primary and foreign keys.
- Insert a sufficient number of sample records.
- Execute the required SQL queries.
- Capture or export the query results.
- Document each query and explain the business purpose of the result.

13. VALIDATION AND TESTING

Every SQL query should be tested against expected results. The student should verify that filtering and sorting conditions produce the intended output.

- Check that SELECT returns the requested columns.
- Verify DISTINCT removes duplicate values.
- Test WHERE conditions using records that both satisfy and fail the condition.
- Verify ASC and DESC sorting.
- Test AND and OR combinations separately.
- Check price-range queries at boundary values.
- Verify that unavailable and out-of-stock products are handled correctly.
- Validate aggregate results against the underlying records.

Example test cases

Test	Expected behaviour
price < 2000	Only products below ■2,000
stock_quantity > 0	Only products currently in stock
availability = 'Available'	Only available products
price BETWEEN 1000 AND 3000	Products within inclusive price range
ORDER BY price DESC	Highest price appears first
SELECT DISTINCT city	Each city appears once

14. FINAL REPORT FORMAT

The final report should contain the SQL query, a short explanation of its purpose, and the resulting output. Screenshots from the SQL environment can be included where required.

15. FINAL PROJECT DELIVERABLES

The completed Week-7 project should contain:

1. E-Commerce database schema.
2. Customers, Products, Categories, Orders, and Order_Items tables or an equivalent documented schema.
3. Sample data for testing.
4. SELECT queries for customer and product information.
5. WHERE queries for filtering records.
6. ORDER BY queries for ascending and descending sorting.
7. DISTINCT queries for unique values.
8. Product search queries based on price, category, and availability.
9. Combined filtering conditions using AND and OR.
10. Basic aggregate business reports using COUNT, AVG, MIN, and MAX.
11. Query validation and test cases.
12. Screenshots or exported results of important queries.
13. Short explanation of the business purpose of each major query.

PROJECT OBJECTIVE

The main objective of this project is to develop practical SQL skills by working with an E-Commerce database. Students will use SELECT, WHERE, ORDER BY, DISTINCT, filtering conditions, and basic aggregate functions to retrieve information and generate useful business reports. The project connects fundamental SQL concepts with realistic online-shopping data.

FINAL OUTPUT

Output	Description
Customer Information	Filtered customer details
Product Information	Product names, prices, categories, and stock
Product Search	Search by price, category, and availability
Filtered Results	Results based on specified conditions
Sorted Results	Ascending or descending business lists
Business Reports	Basic counts, averages, minimums, and maximums

Note: The sample SQL queries and sample records in this document are illustrative. The final query results should be generated from the actual E-Commerce database created for the project.